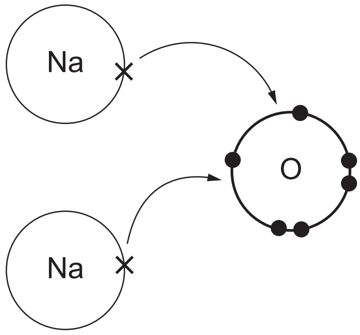
Examiner only

3. (a) The diagram shows the transfer of electrons that takes place during the formation of sodium oxide.



(i) Name the type of bonding present in sodium oxide. [1]

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(ii) State what must be done to sodium oxide so that it will conduct electricity.

Explain your answer. [2]

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Examiner only

(b) (i) Draw a dot and cross diagram to show the bonding in a molecule of tetrafluoromethane, CF_4. [2]

carbon (C) 2,4 fluorine (F) 2,7

(ii) Tetrafluoromethane is a simple covalent substance and is a gas at room temperature.

Explain why it has a low boiling point. [2]

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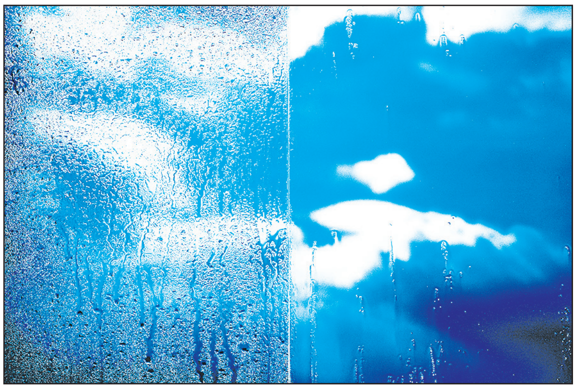
(c) Nanoparticles are extremely small particles that have different properties to the same material at bulk size.

Nano-scale titanium dioxide is commonly used in sunscreen and in self-cleaning windows.



Give the reason why nano-scale titanium dioxide is effective in each of these uses. [2]

Sunscreen

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Self-cleaning windows

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